Story space: salient concepts as volumes in a story-conditioned latent space

1 Problem

Definition 1 (Story, paragraphs)**.** *A story $s \in \mathcal{T}$ is a string of text from the space of texts $\mathcal{T}$. It is split into paragraphs $p_1, \ldots, p_N \in \mathcal{T}$.*

s	the story, a string of text
$\mathcal{T}$	the space of all strings of text
p_i	the story's i-th paragraph
N	the number of paragraphs in the story

Definition 2 (Salient concept)**.** *A salient concept $c \in \mathcal{T}$ is a string of text naming an idea the story contains. It need not be copied out of the story: the idea may never be mentioned directly. A story comes with a set of salient concepts $C_s = \{c_1, \ldots, c_K\}$.*

c	a salient concept, a string of text naming an idea
C_s	the set of the story's salient concepts
K	the number of salient concepts

Definition 3 (Story state)**.** *A story state σ is a structured description of a moment of the story (characters, location, event, action). A fixed template τ renders it as text, $\tau(\sigma) \in \mathcal{T}$.*

σ	a story state: characters, location, event, action
τ	the fixed template that writes a state out as text
$\tau(\sigma)$	the text of state σ

Definition 4 (Story space)**.** *The story space $\mathcal{Z}_s \subseteq \mathbb{R}^d$ is the latent space of story s. Every salient concept c is a volume $B_c \subset \mathcal{Z}_s$ and every story state σ is a point $z(\sigma) \in \mathcal{Z}_s$. A state is related to a concept when its point lies inside the concept's volume. Two concepts are related when their volumes overlap.*

$\mathcal{Z}_s$	the story space of story s
d	the number of dimensions of the story space
$\mathbb{R}^d$	the space of vectors of d real numbers
B_c	the volume of concept c in the story space
$z(\sigma)$	the point of state σ in the story space

Figure 1 shows two overlapping concept volumes and a story moving through them.

The story space is a conditioning of the general latent space of all texts: of all the places a text can occupy in the general space, the story picks out the region that it, its concepts and its states occupy.

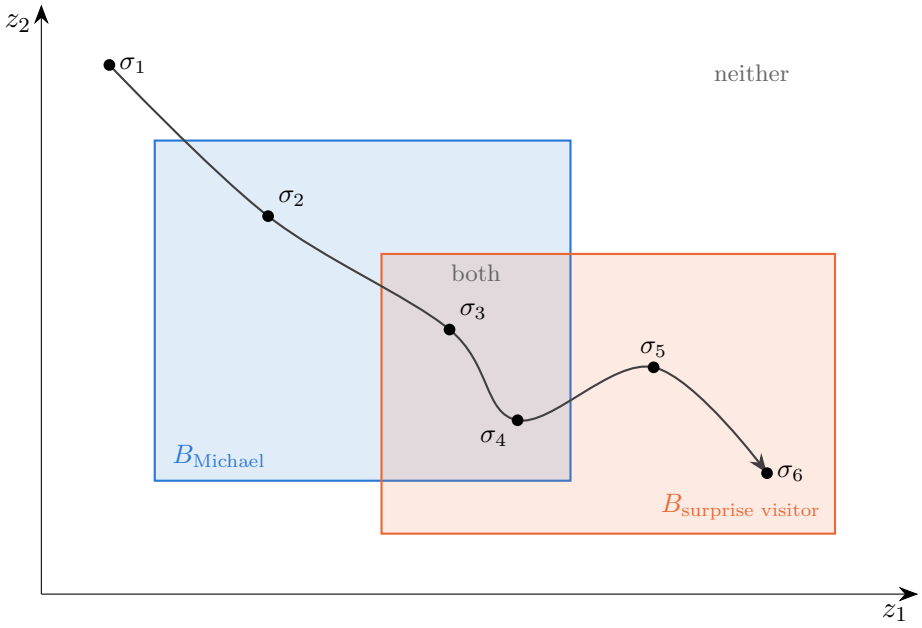


Figure 1: Illustration, not computed from data. Two concept volumes in a story space and the trajectory of SCENE_1's states in story order. z_1 and z_2 are two of the story space's d dimensions; σ_k marks the point of the story's k-th state. σ_1, a warm sunny day, lies in neither volume. σ_2, Michael and Sarah setting out for a walk, lies only in B_{Michael}. σ_3, an unexpected knock at their door, and σ_4, Michael going down to check, lie in the overlap. σ_5, Sarah calling "Who is it?", and σ_6, no answer coming, lie only in $B_{\text{surprise visitor}}$.

Tasks.

(T1) Define the story space for s, given s and C_s: the operation `encode_basis`.

(T2) Map a story state into the story space: the operation `encode_from_state`. States are described with arbitrary text, yet two states described with the same words must land on the same point.

(T3) Find the boundaries of the volumes of the salient concepts.

Requirements.

Requirement 1 (Non-degenerate volume)**.** *The volume of a concept has non-zero width along every dimension of $\mathcal{Z}_s$.*

Requirement 2 (Containment)**.** *A state related to concept c maps to a point inside B_c. A state unrelated to c maps outside it.*

Requirement 3 (Multiple membership)**.** *A state may be related to several concepts at once, so volumes may overlap and a point may lie in several of them.*

Requirement 4 (Identity)**.** *Equal states map to the same point: $\sigma = \sigma' \Rightarrow z(\sigma) = z(\sigma')$.*

Requirement 5 (Separation)**.** *Different states map to different points: $\sigma \neq \sigma' \Rightarrow z(\sigma) \neq z(\sigma')$.*

σ' a second story state, compared with σ

2 How Qwen encodes a text

Qwen is used as it is: frozen, nothing in it trained. The figures were measured with Qwen3-8B, which has $L = 36$ layers and a latent space of $D = 4096$ dimensions.

A point in latent space. Qwen splits a text t into tokens $k = 0, \ldots, n-1$. After each layer ℓ it holds, for every token, a point $h_{\ell,k}(t)$ in its latent space $\mathbb{R}^D$; $\ell = 0$ is the token's lookup before any layer.

The latent space. $\mathbb{R}^D$ is a real vector space with the dot product $u \cdot v = \sum_{i=1}^{D} u_i v_i$, which gives lengths, angles and distances. All layers share it. The points are not normalised, and their lengths are bounded: the longest here is about 12,000. The points crowd into a narrow cone whose apex is the space's zero, because training gives every point one large common component.

One point per token. Qwen gives one point per token, not one per text. Each point summarises the text up to and including its token. The text "Bob" gives one point after each layer; a passage of n tokens gives n.

A token's path through the layers. Following one token from layer 0 to layer L gives a trajectory of $L + 1$ points. Each step is one layer: the layer analyses the point, attending to the points of the tokens before it and passing the result through a feed-forward network, and adds what it finds to the point:

$$h_{\ell,k}(t) = h_{\ell-1,k}(t) + \Delta_{\ell,k}(t). \tag{1}$$

Where the information is. In each point itself: in where it lies along the space's feature directions, not in how the points relate to each other.

Token 0**.** From layer 7 on, the first token's point becomes an attention sink: about 185 times longer than the others and the same for every text, whatever its first word. It is left out of everything below.

t	any text given to Qwen
k	a token's position in the text, starting at 0
n	the number of tokens in the text
ℓ	a layer of Qwen; 0 means the lookup before the first layer
L	the number of Qwen's layers
D	the number of dimensions of Qwen's latent space
$\mathbb{R}^D$	Qwen's latent space
$h_{\ell,k}(t)$	the point in $\mathbb{R}^D$ that Qwen holds for token k of text t after layer ℓ
u, v	any two points of $\mathbb{R}^D$
u_i	the i-th of the D coordinates of u
$u \cdot v$	the dot product of u and v
$\Delta_{\ell,k}(t)$	what layer ℓ adds to token k's point

2.1 Extending a text

Three texts share a beginning:

A	"Bob buys Alice coffee"
A + B	"Bob buys Alice coffee, because he loves her"
A + C	"Bob buys Alice coffee and bagels with salmon"

B adds Bob's motivation; C adds a detail that changes nothing important. Figure 2 shows their points after layer 16.

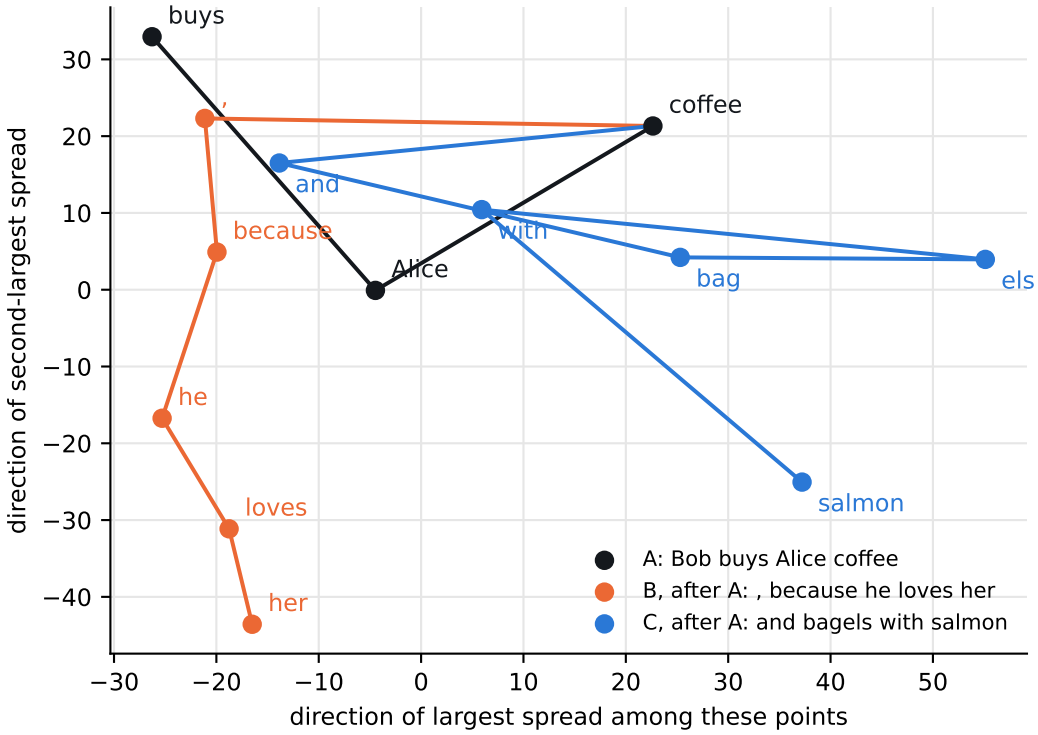


Figure 2: The points of A, A + B and A + C after layer 16, token 0 ("Bob") left out. Black: A's points, shared by all three texts. Orange: the points B adds after A. Blue: the points C adds after A. Lines join the points in reading order. The 13 points live in $\mathbb{R}^{4096}$; they are drawn along the two directions in which they differ most, which hold 34% of their spread.

Salient information. Write $I(t)$ for the salient information a text t carries. We take as given:

$$I(A) \approx I(B), \qquad I(A) > I(C), \qquad I(A+B) \gg I(A+C), \qquad I(A) \approx I(A+C). \tag{2}$$

What remains to be found is how Qwen encodes and represents this information.

$I(t)$ the salient information text t carries

Shannon information. The Shannon information of a text under Qwen is the number of bits Qwen needs to predict it, token by token:

$$H(t) = -\sum_{k} \log_2 p\big(\text{token}_k \mid \text{tokens before it}\big), \qquad H(B \mid A) = H(A+B) - H(A). \tag{3}$$

$H(t)$ the Shannon information of text t under Qwen, in bits
$p(\text{token}_k \mid \cdot)$ the probability Qwen gives token k after reading the tokens before it
$H(B \mid A)$ the bits B costs once A has been read

Every token is scored, the first one included, by starting each text with Qwen's document separator.

text	H (bits)
A	63.4
B	50.8
C	72.6
A + B	92.1
A + C	111.9
B, after A: $H(B \mid A)$	28.7
C, after A: $H(C \mid A)$	48.6

Table 1: The Shannon information of every text under Qwen3-8B.

salient information	Shannon information	holds
$I(A) \approx I(B)$	$H(A) = 63.4$, $H(B) = 50.8$	no: A costs a quarter more
$I(A) > I(C)$	$H(A) = 63.4$, $H(C) = 72.6$	no: C costs more
$I(A+B) \gg I(A+C)$	$H(A+B) = 92.1$, $H(A+C) = 111.9$	no: the reverse
$I(A) \approx I(A+C)$	$H(A) = 63.4$, $H(A+C) = 111.9$	no: A + C costs 76% more

Table 2: The relations of salient information, tested with Shannon information.

None of the four relations holds for Shannon information (Tables 1 to 2). Every added token costs bits, so a longer text always carries more, and what costs most is what surprises Qwen: "salmon" costs 16.4 bits, "her" in "because he loves her" 0.4. Salient information is therefore not what Qwen's surprise measures.

2.2 Properties of the function that measures salient information

The data give the function I three properties:

1. **It is not surprise.** Shannon information breaks all four relations (Table 2), so the function cannot be built from Qwen's token probabilities.

2. **It is not additive over tokens.** $I(A) \approx I(A+C)$ although A + C has five more tokens than A, so the function must give C's points almost nothing while giving B's a lot.

3. **It is not distance.** B and C move the points by about the same amount: after layer 16, the last points of A + B and A + C lie equally far from A's last point (cosine 0.34 and 0.37, distance 96 and 95). They move them in different directions (Figure 2), so the function must tell directions apart.

Together: the function is a projection onto a subspace of Qwen's latent space that keeps the directions B's points move in and discards the directions C's points move in.

2.3 Movement

Figure 3 shows the points of A, B and C as areas with their centroids after layers 1, 10, 20 and 36.

Figure 4 runs A, B, C, A + B and A + C through Qwen separately and draws the area of each.

Adding B or C never moves A's points: inside A + B and A + C they stay within 2.4% of their length of where A alone puts them, which is rounding. Qwen reads left to right, so A's tokens cannot see what follows them. B's and C's tokens do see A, and A moves their points away from where they sit alone, by a share of their length that grows with depth and then falls:

after layer	1	10	16	20	36
distance from where they sit alone	16–29%	51–86%	64–84%	59–81%	24–59%

Everything A + B means beyond A is therefore written into B's points. The areas in Figures 3 and 4 do not represent the meaning of a sentence: an area only shows how spread out one text's token points are. What reading A does to B is the drift: the gap between where B's points sit alone and where they sit inside A + B.

The areas do not measure salient information either:

1. **Their size does not grow with salient information.** From layer 10 on, A + C's points are more spread out than A + B's, although $I(A+B) \gg I(A+C)$.

2. **Their centroids do not tell B from C by distance.** A + B's and A + C's centroids lie equally far from A's centroid. They lie in different directions from it, as the last points do in the properties above, so the two centroids are as far from each other as from A's.

after layer	1	10	16	20	36
spread of A + B	10.2	43.4	54.5	71.0	59.6
spread of A + C	10.2	47.6	58.8	75.1	67.7
distance of A + B's centroid from A's	6.3	25.1	31.9	42.0	32.5
distance of A + C's centroid from A's	6.1	25.8	31.6	40.2	30.0
distance between A + B's and A + C's centroids	3.7	23.5	30.3	39.6	30.7

The spread of a text is the root mean square distance of its points from their centroid, measured in $\mathbb{R}^{4096}$, token 0 left out.

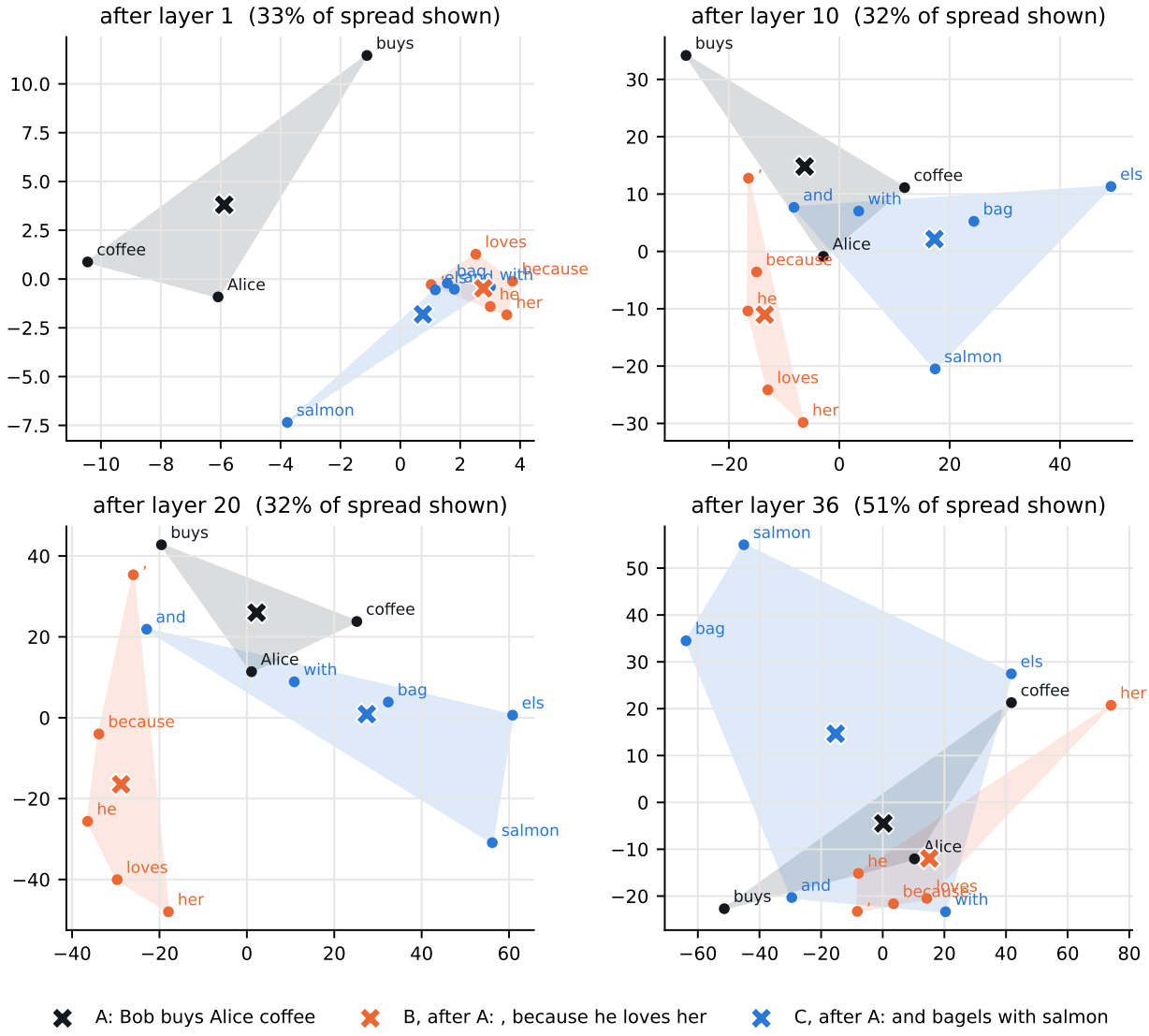


Figure 3: The points of A, A + B and A + C after layers 1, 10, 20 and 36, token 0 ("Bob") left out. Black: A's points. Orange: the points B adds after A. Blue: the points C adds after A. Each shaded area is the smallest polygon around one colour's points; the cross is their centroid. Each panel draws its layer's 13 points along the two directions in which they differ most, with the share of their spread those directions hold in the title.

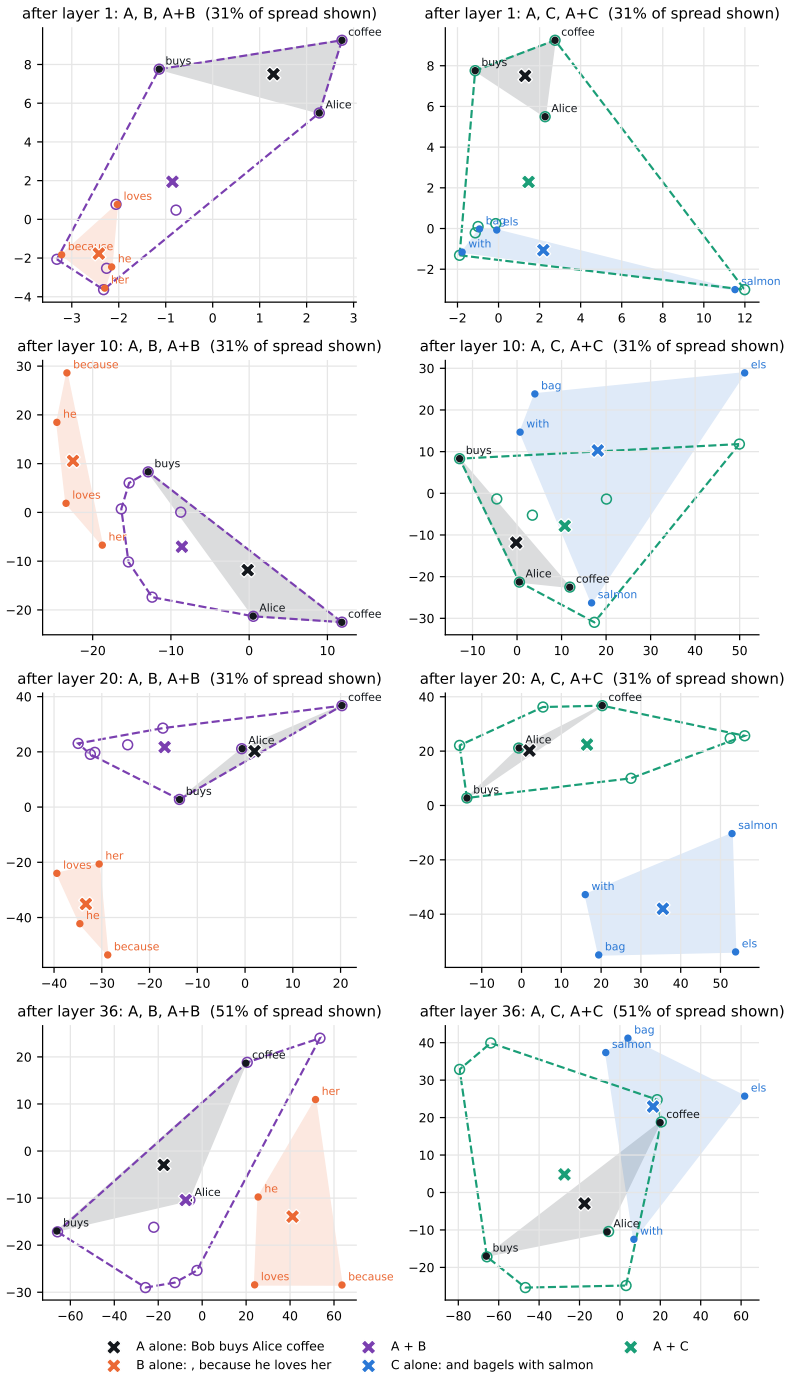


Figure 4: A, B, C, A + B and A + C, each run through Qwen on its own, after layers 1, 10, 20 and 36, every text's token 0 left out. Filled areas: A, B and C alone. Dashed outlines and hollow circles: A + B (left) and A + C (right). Crosses: centroids. Both panels of a row are drawn along the same two directions, those in which that layer's points differ most.

2.4 Volumes that represent salient information

Figures 3 and 4 show the transformations Qwen applies to tokens: they project them into a generic latent space, shaped by the loss Qwen was trained with, next-token prediction. We want to fine-tune or adapt Qwen's output so that volumes begin to represent salient information. $V(t)$ is the volume of text t in the adapted space.

$V(t)$	the volume of text t in the adapted space
$\operatorname{vol} V(t)$	the size of that volume
$V(t) \cap V(t')$	the overlap of the volumes of texts t and t'

Requirement 6 (Size is salience)**.** *The larger the salient information, the larger the volume:* $I(t) > I(t') \implies \operatorname{vol} V(t) > \operatorname{vol} V(t')$.

Requirement 7 (Overlap is shared salience)**.** *Volumes that overlap share salient information:* $V(t) \cap V(t')$ *is the salient information* t *and* t' *share.*

For example, with four texts

A	"Bob buys Alice coffee"
B	"because he loves her"
D	"Charlie takes Jean to the movies"
E	"where they share an intimate kiss"

the volumes must satisfy

$$\begin{aligned} V(A) \cap V(D) &= \emptyset, \\ V(A+B) \cap V(D+B) &= V(B), \\ V(A+B) \cap V(D+E) &= V(B) \cap V(E) \neq \emptyset, \\ V(A+E) \cap V(D+B) &= V(B) \cap V(E) \neq \emptyset. \end{aligned}$$

Figures 5 and 6 illustrate how these relations would look on transformed points.

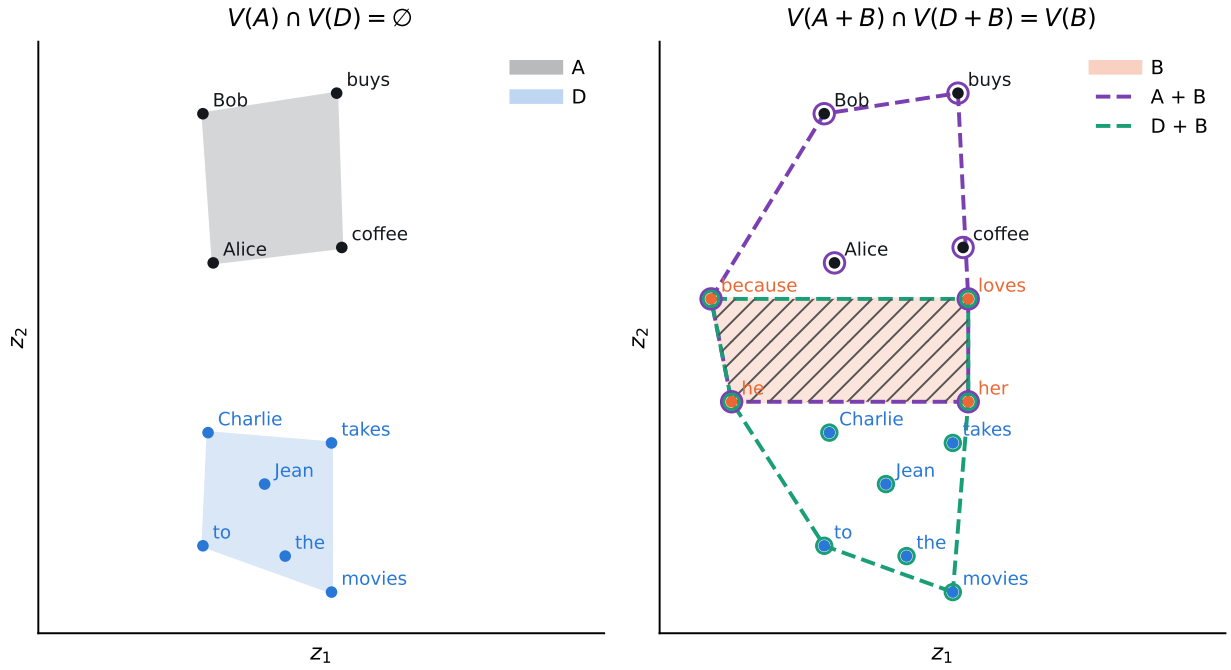


Figure 5: Illustration, not computed from data: the first two relations on transformed points. Each dot is the point of one token in the adapted space; z_1 and z_2 are two of its dimensions. A text's volume is the smallest polygon around its tokens' points. Filled areas: texts alone. Dashed outlines and hollow circles: joined texts, whose points are the points of their parts. Hatched: the overlap of the two dashed volumes, the left side of the relation. Left: $V(A) \cap V(D) = \emptyset$; A's and D's volumes do not meet. Right: $V(A + B) \cap V(D + B) = V(B)$; A + B and D + B overlap exactly in B's volume.

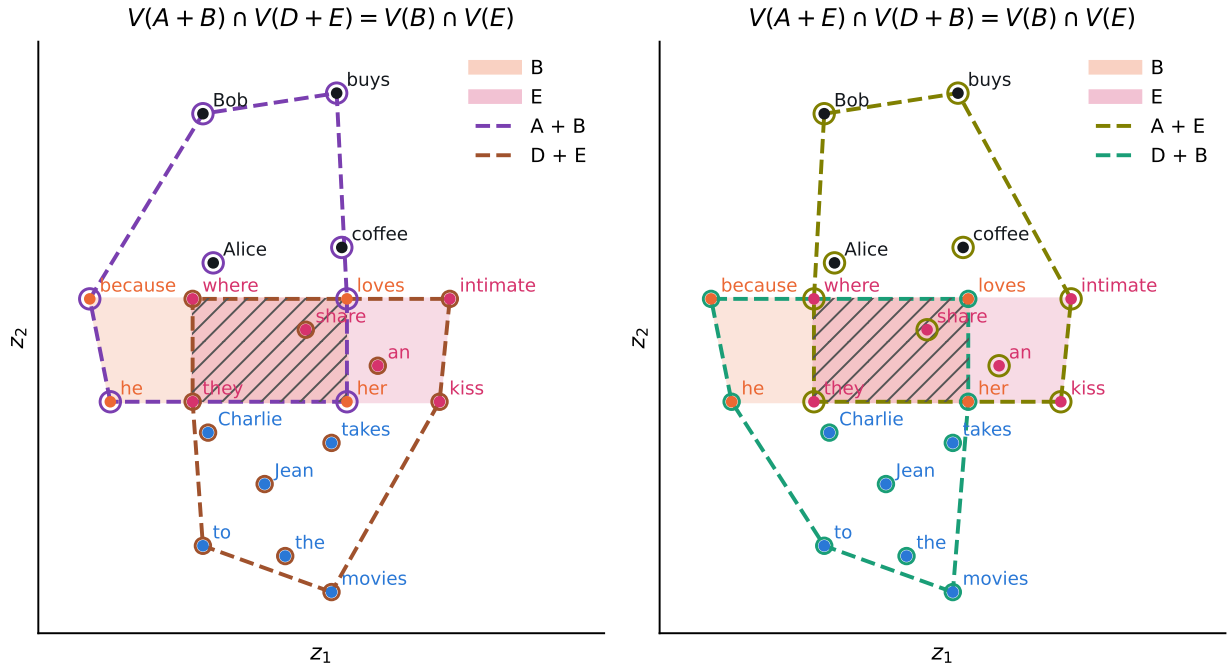


Figure 6: Illustration, not computed from data: the last two relations on transformed points, drawn as in Figure 5. B and E overlap. Left: $V(A+B) \cap V(D+E) = V(B) \cap V(E)$; A + B and D + E overlap exactly where B and E do. Right: $V(A+E) \cap V(D+B) = V(B) \cap V(E)$; so do A + E and D + B.